

Coding Theory and Mathematics

– Linear, Hamming, and Cyclic Codes –

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수학의 즐거움, Enjoying Math

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Contents



I. Introduction

I. Introduction

I. Introduction

Errors Are Everywhere



Physical world

- ▶ Scratches on a CD or DVD

Digital world

- ▶ Bit flips during internet transmission



Questions.

“How do we transmit (or store) data **reliably** in the presence of noise?”

I. Introduction

Can We Make Noisy Data Reliable Again?



...1 1 1 1 0 1 1 0 1 0 1 0 1 1 0 0 1...

↓ noise

...1 1 1 1 0 1 1 0 1 0 1 0 **0** 1 0 0 1...

I. Introduction

Can We Make Noisy Data Reliable Again?



...1 1 1 1 0 1 1 0 1 0 1 0 1 1 0 0 1...

↓ noise

...1 1 1 1 0 1 1 0 1 0 1 0 **0** 1 0 0 1...

► Can we **detect** it?

I. Introduction

Can We Make Noisy Data Reliable Again?



...1 1 1 1 0 1 1 0 1 0 1 0 1 1 0 0 1...

↓ noise

...1 1 1 1 0 1 1 0 1 0 1 0 **0** 1 0 0 1...

► Can we **detect** it?

► Can we **fix** it?

I. Introduction

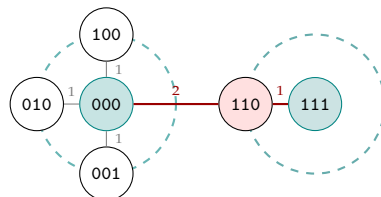
Triple Repetition Code $[3,1]_2$



The simplest way to correct errors: **say it three times.**

$$0 \mapsto 000, \quad 1 \mapsto 111.$$

Received	Majority	Decision
000	0×3	0
001	0×2	0
010	0×2	0
100	0×2	0
011	1×2	1
101	1×2	1
110	1×2	1
111	1×3	1



2-bit errors are detectable, but not always correctable.

Performance

- Detects any two-bit flips
- Corrects any single-bit flip

Cost

- Rate $R = 1/3$: send 3 bits to convey 1 bit
- Very inefficient!

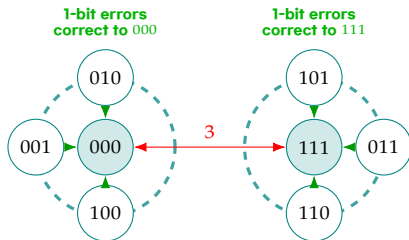
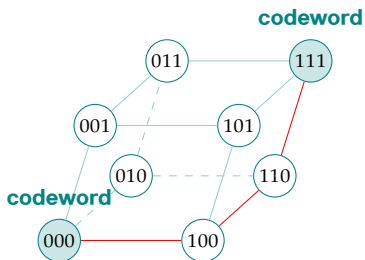
Question. Can we do much better?

I. Introduction

The Key Idea: Distance



Why does the triple repetition code work? The two codewords are **far apart**.



Minimum distance

A code with minimum distance d corrects all errors of weight $\leq \left\lfloor \frac{(d-1)}{2} \right\rfloor$.

I. Introduction

Hamming Distance



Hamming distance counts how many coordinates differ. For any $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$,

$$d_H(\mathbf{x}, \mathbf{y}) = w_H(\mathbf{x} \oplus \mathbf{y}) = \#\{i : x_i \neq y_i\}$$

where w_H is Hamming weight (number of ones).

Example 1. $d_H(0101, 1100) = 2$.

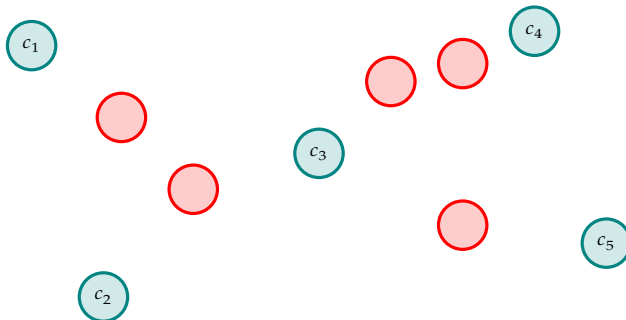
Example 2. $d_H(000, 101) = 2$.

Hamming distance is a metric:

- (i) $d_H(\mathbf{x}, \mathbf{y}) \geq 0$, $d_H(\mathbf{x}, \mathbf{y}) = 0 \iff \mathbf{x} = \mathbf{y}$,
- (ii) $d_H(\mathbf{x}, \mathbf{y}) = d_H(\mathbf{y}, \mathbf{x})$,
- (iii) $d_H(\mathbf{x}, \mathbf{z}) \leq d_H(\mathbf{x}, \mathbf{y}) + d_H(\mathbf{y}, \mathbf{z})$.

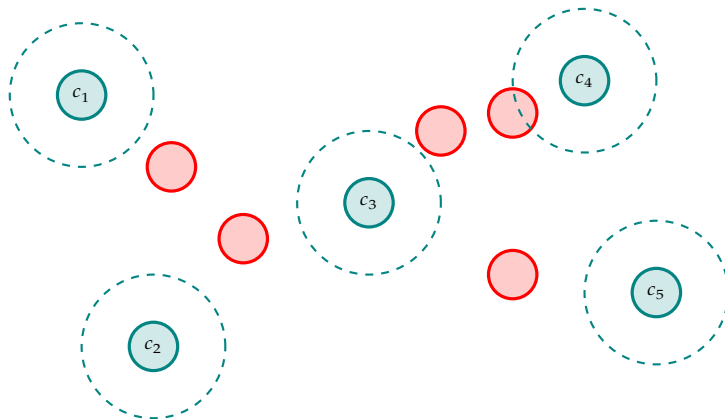
I. Introduction

Correction Balls



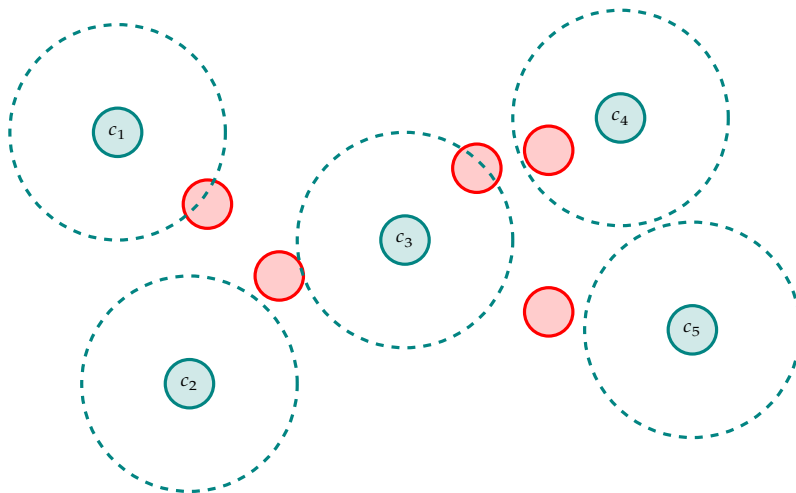
I. Introduction

Correction Balls



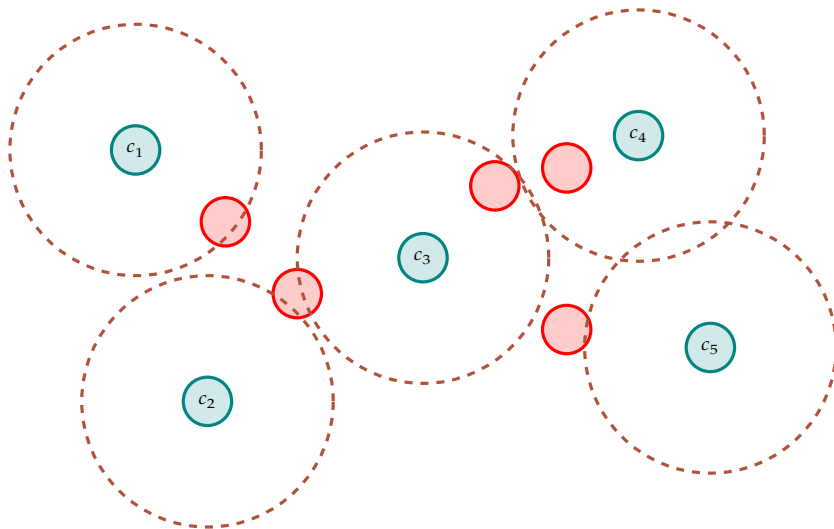
I. Introduction

Correction Balls



I. Introduction

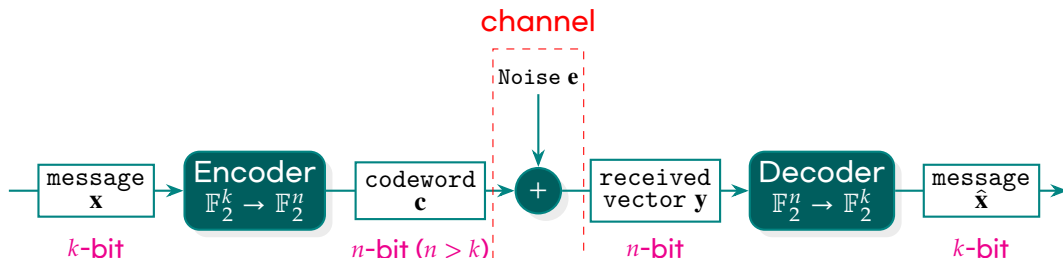
Correction Balls



Remark. No overlap of correction balls $\Leftrightarrow$ unique decoding is possible.

I. Introduction

The Communication Model



Setup

- ▶ Message $\mathbf{x} \in \mathbb{F}_q^k$
- ▶ Encoder : $\mathbf{x} \mapsto \mathbf{c} \in \mathbb{F}_q^n$, $n > k$
- ▶ Channel : $\mathbf{y} = \mathbf{c} \oplus \mathbf{e}$

Goal

- ▶ Recover $\hat{\mathbf{x}} = \mathbf{x}$ even when $\mathbf{e} \neq \mathbf{0}$
- ▶ Keep rate $R = k/n$ as large as possible

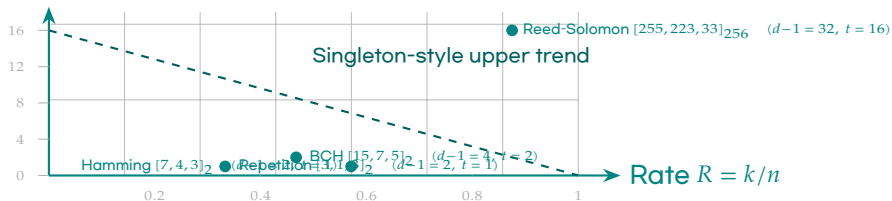
I. Introduction

The Fundamental Tradeoff



Code	Rate	Detects	Corrects
$[n, k, d]_q$ -code	k/n	$d - 1$	$\lfloor (d - 1)/2 \rfloor$
Repetition $[3, 1, 3]_2$	$1/3 \approx 0.33$	2	1
Hamming $[7, 4, 3]_2$	$4/7 \approx 0.57$	2	1
BCH $[15, 7, 5]_2$	$7/15 \approx 0.47$	4	2
Reed-Solomon $[255, 223, 33]_{256}$	$223/255 \approx 0.87$	32	16

Correctable errors



Detects (from table values)



I. Introduction

Shannon's Channel Coding Theorem (1948)



Shannon's Channel Coding Theorem

For any discrete memoryless channel with capacity C and any rate $R < C$, there exists a sequence of codes of rate $\rightarrow R$ whose block error probability $\rightarrow 0$.

What it guarantees

- ▶ Reliable communication is possible below capacity
- ▶ Error probability can be made arbitrarily small
- ▶ A fundamental existence result

What it does not provide

- ▶ How to construct such codes explicitly
- ▶ How to decode efficiently
- ▶ The proof uses random codes!

The Central Challenge of Coding Theory

Construct **explicit algebraic codes** that are both **efficient** (high rate) and **powerful** (correct many errors).

I. Introduction

Shannon's Theorem — Plain Language



Analogy. Imagine shouting across a noisy cafeteria.

- ▶ The capacity C is a property of the cafeteria — how noisy it is.
- ▶ If you speak below capacity ($R < C$), there exists a way of encoding your message so that the other person can always understand you — no matter how noisy.
- ▶ But Shannon doesn't tell you what to say. He only guarantees that the right words exist.

- 1948: Shannon's theorem
- 1950: Hamming codes
- 1954: Reed-Muller codes
- 1960: Reed-Solomon codes
- 1993: Turbo codes
- 1995: LDPC (re-discovered)
- 2009: Polar codes

The punchline

Algebra gives us the explicit construction. Linear codes, Hamming codes, cyclic codes, Reed-Solomon codes — these are Shannon's theorem made concrete.

I. Introduction

Three Beautiful Families

Each family adds a layer of algebraic structure:

Linear Codes

subspaces of $\mathbb{F}_q^n$

▼ more structure

Hamming Codes

projective geometry

▼ more structure

Cyclic Codes

polynomial rings

II. Linear Codes

- Subspaces of $\mathbb{F}_q^n$; generator matrix G , parity-check H
- Key bounds: Hamming, Singleton, Griesmer

III. Hamming Codes

- Columns of H = all nonzero points of $\text{PG}(r-1, q)$
- Perfect single-error-correcting codes

IV. Cyclic Codes

- Ideals in $\mathbb{F}_q[X]/(X^n - 1)$;



I. Introduction

Roadmap



II. Linear Codes

Structure via **linear algebra**

- Subspaces of $\mathbb{F}_q^n$
- Generator G , parity-check H
- Hamming distance & error correction

III. Hamming Codes

Perfect codes via **projective geometry**

- Parity checks from projective points
- Syndrome decoding as geometry

IV. Cyclic Codes

Structure via **polynomial rings**

- Quotient $\mathbb{F}_q[X]/(X^n - 1)$
- Ideals and generator polynomials
- BCH design and cyclic Hamming codes

Key Observation

Coding theory is a meeting point of **linear algebra**, **finite geometry**, and **polynomial algebra**.

Thank You

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